



Microsoft

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IT SOLUTIONS & *TECHNOLOGY*

By: Denita Jefferson, Tyler Pettiford & Ejaaz Kerem



Microsoft

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Meet the Team



Denita Jefferson
*AI Digital Operations
Center*



Tyler Pettiford
*Phishing Detection
Agent*



Ejaaz Kerem
*Call to Architecture
Agent*

About Me

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Tyler Pettiford | Microsoft On Ramps Program



Who I am

A graduate of McKinley Technology High School, now a full-time student at Morehouse College.

What drives me

Curiosity and innovation. I love learning how things work, and I'm passionate about building the laws, tools, and platforms that improve society.

Outside of work

Gaming has been part of my life for as long as I can remember, and it's a big reason I chose to pursue computer science as a career.



AUTONOMOUS EMAIL DEFENSE

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Phantom

AI Agent

An AI security analyst that reads your inbox and handles the busywork, **spotting email threats and taking on everyday cyber tasks on demand.**

01

SCAN

YOUR INBOX

02

SPOT

EMAIL THREATS

03

ACT

ON DEMAND

How Phantom hunts phishing

Threat detection built into every message you receive

01 Phantom Reviews Requested Email

Once granted access and permission, Phantom scans sender, headers, links, and language the moment mail arrives

02 Flags Malicious Content

Detects spoofed domains, look-alike URLs, urgent-payment lures, and credential-harvesting pages.

03 Explains the verdict

Returns a clear risk score with the exact reasons, so users know why a message was held.

04 Acts on your behalf

If prompted, Phantom can delete and/or report the flagged email on your behalf

Phantom Demo

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A live verdict on a suspicious invoice email

Phantom | phishing verdict

Hi, Can you take a quick look at the attached invoice when you have a chance? I think there may be an issue with the formatting because some of the HTML appears to have been copied into the message instead of rendering correctly. The text below is part of the broken template:

```
<script>alert('XSS Test')</script>
```

I'm not sure why it showed up like this, but it might help your IT team troubleshoot the problem. Thanks, Tyler

Phantom

Yes — this email is **suspicious**.

Red flags

- **Invoice lure:** unexpected invoice/payment themes are common phishing bait.
- **Broken HTML explanation:** “formatting issue” can be used to make malicious content seem harmless.
- **Embedded script text:** `<script>alert('XSS Test')</script>` is a strong indicator of either:
 - a **template/security test**, or
 - a **malicious/suspicious HTML payload reference**.
- **Pressure to open an attachment:** “take a quick look at the attached invoice” is classic phishing language.
- **Generic sign-off:** only “Thanks, Tyler” with no business context.

Assessment

- **Risk level:** **Medium to High**
- **Most likely threats:** phishing, malicious attachment, or HTML/script-based abuse attempt.
- If the script appeared literally as escaped text, it may not execute by itself, but it is still **not normal for a legitimate invoice email**.

What to do

1. **Do not open the attachment.**
2. **Do not reply** or click any links.
3. Verify with the supposed sender using a **known-good contact method**.
4. Report it to your **IT/security team** as a suspicious invoice email.
5. If you want, paste the **headers, sender address, attachment name/type, and any links** and I'll do a deeper triage.

If you want, I can also turn this into a short **phishing analysis report** for your IT team.

What Phantom found

RISK LEVEL: HIGH

- **Verdict:** Likely phishing or a malicious lure
- **Bait:** An attached invoice, with script code hidden in the message body
- **Tell:** Generic sign-off, no company details or invoice number
- **Guidance:** Do not open or click. Report and quarantine the message



Summer Recap

Overall, Microsoft is a phenomenal company that fosters a welcoming, collaborative, and productive work environment, making it an excellent place for employees to grow both personally and professionally.

What I enjoyed most

- Meeting new talented individuals
- Studying for the SC-900
- The in-person office experience
- Assisting peers with problems I may have already faced
- Working on researching and developing my own AI agent

ABOUT ME

Ejaaz Kerem | Microsoft On Ramps Program





Call to Architecture Agent

I noticed newer CSAs take way longer to build the judgment senior architects have instinctively, so I built an AI agent that closes that gap.

—



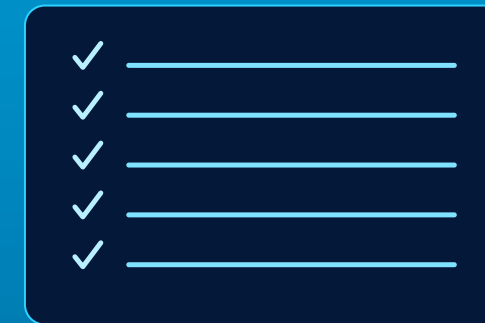
THE GAP

- Transcripts aren't requirements
- WAF judgment takes years
- Knowledge lives in people's heads



Transcript

≠



Requirements



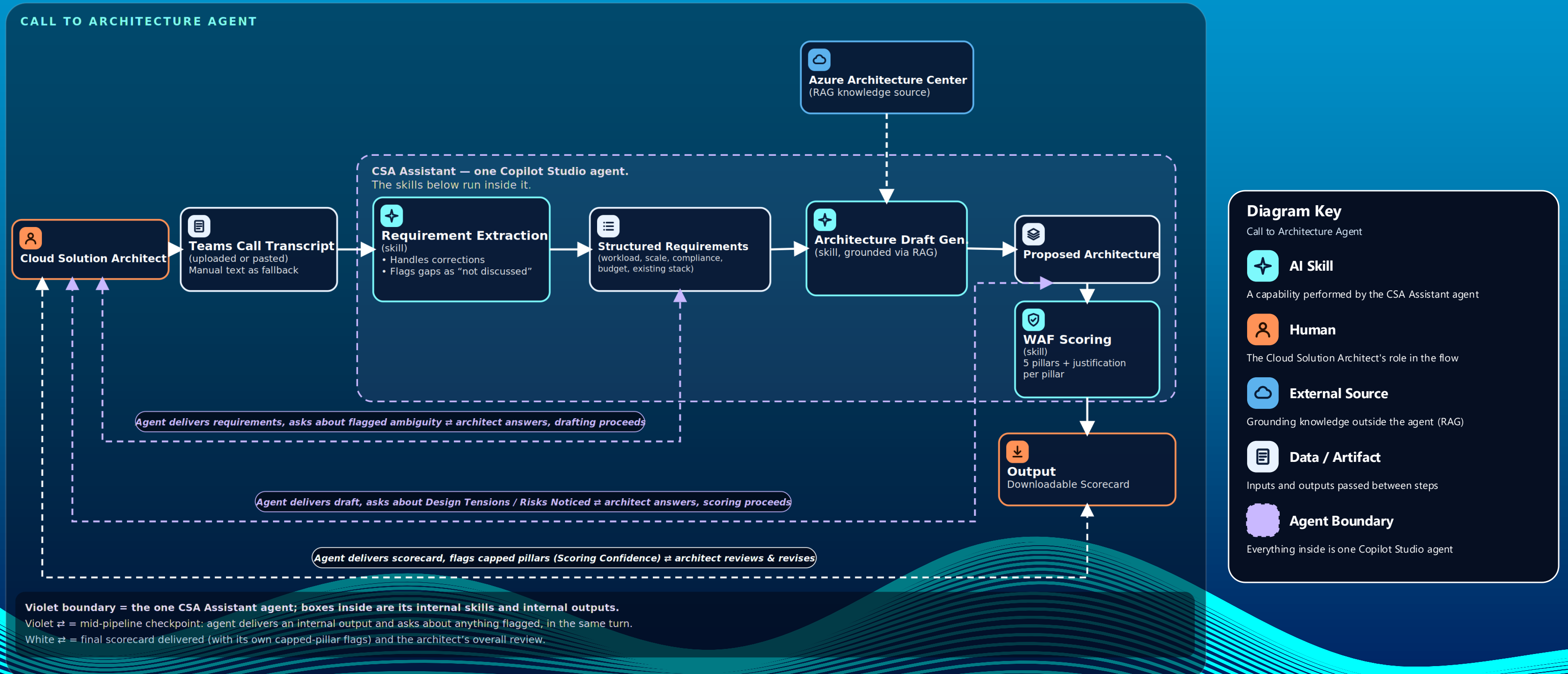
New CSA



Senior CSA



How it works





Microsoft

DEMO

b-ejaazkerem@microsoft.com

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regional clinics coming online, it's actually going to be closer to 9 million records a year, not 2 million.

0:01:18 Alex Kim: Budget-wise, we've already got some reserved Azure credits left over from last year we're trying to use up. We're not looking to take on a lot of new ongoing usage-based spend if we can help it.

0:01:33 Dana Reyes: Anything else before we wrap?

0:01:36 Alex Kim: Don't think so, we can circle back on reporting cadence later.

CSA Assistant

- The user has provided a customer call transcript. I should use the Extract Requirements skill.
- Loaded Skill: Default_CSA_Requirements_Extractor_322cd9.skill.extract_5ei >
- Let me read the transcript carefully and extract requirements following the skill instructions. **

Ask a question or describe what you need



AI-generated content may be incorrect. [Learn more](#)

CSA First-Draft Architecture Package

Workload: Retail Order Platform Migration

Prepared:

This is a first-draft output from the CSA Assistant, generated from a customer call transcript. It has not been reviewed for accuracy or completeness — treat every service choice, tag, and open item below as a starting point for architect review, not a final design.

Requirements Summary

Workload Type

Customer-facing retail order platform (.NET web application) migrating from on-premises servers to the cloud. The back-end SQL Server database is not moving to the cloud at this time — only the web application layer is in scope for the migration. "We just want the customer-facing app in the cloud." — Sarah Chen

Expected Scale

[Corrected] Originally stated as ~10,000 daily active users; corrected mid-call to up to 100,000 daily active users during peak periods (driven by planned seasonal sale events) using 100,000. "Traffic's going to be a lot higher than 10,000 — marketing's planning big seasonal sale events, so we should design for up to 100,000 daily users during peak periods." — Mike Torres

Compliance Needs

PCI DSS compliance is required. The customer processes card payments directly and explicitly described this as non-negotiable, citing a prior vendor audit. "Since we process card payments directly, we'll need to be PCI DSS compliant. That's non-negotiable given what happened with our last vendor audit." — Sarah Chen

Budget Sensitivity

[Corrected] Initially described as flexible with ~\$50,000/month set aside for infrastructure; walked back mid-call to a leaner, cost-conscious posture due to competing projects. A revised number has not yet been determined — using the corrected signal: budget-constrained, exact figure TBD. "Finance is still working through it, but just know it's not going to be as open as Priya said before." — Sarah Chen

Existing Technology Stack

Web application: .NET (existing, will be deployed to cloud). Database: SQL Server (on-premises, remaining on-prem — not migrating). No other tools, CI/CD systems, or cloud vendors mentioned.

Additional Notes

Stakeholders identified: Sarah Chen (technical/project lead), Mike Torres (engineering/scale concerns), Priya Patel (budget/finance liaison). Timeline: [Gap] Not discussed.

Flags for the Architect

1. Budget figure is unresolved. The \$50k/month figure was stated and then explicitly retracted, but no replacement number was given. 2. Hybrid connectivity gap. The .NET web app will run in Azure but the SQL Server database stays on-prem. No one discussed how the cloud app will connect back to that database.

First-Draft Architecture

Compute

Proposed: Azure App Service (Premium v3 plan) with autoscaling. Requirement tied to: Hosting the existing .NET web application in the cloud; peak traffic of up to 100,000 daily active users driven by seasonal sale events. Reasoning: App Service is the natural fit here — it supports .NET natively, requires no OS-level management or container orchestration expertise, and critically supports both VNet Integration (outbound) and Private Endpoints (inbound), which are



Takeaways

CHALLENGES

- Adapting to the corporate environment
- Learning while building
- Navigating ambiguity and change

LESSONS

- Start with the user not the technology
- Adaptability is essential
- Communication is important as technical skills



ABOUT ME



Denita Jefferson | *Aspiring Cybersecurity Professional*

Recent CS Graduate from Delaware State University

Hobbies: Photography, Painting, Reading, Going on walks

Next Steps: Obtain Cybersecurity certifications

Digital Operations Center

Multi-agent system

Empowering organizations with AI agents to collaborate across IT, security, and business operations to resolve issues, improve efficiency, and enhance visibility.



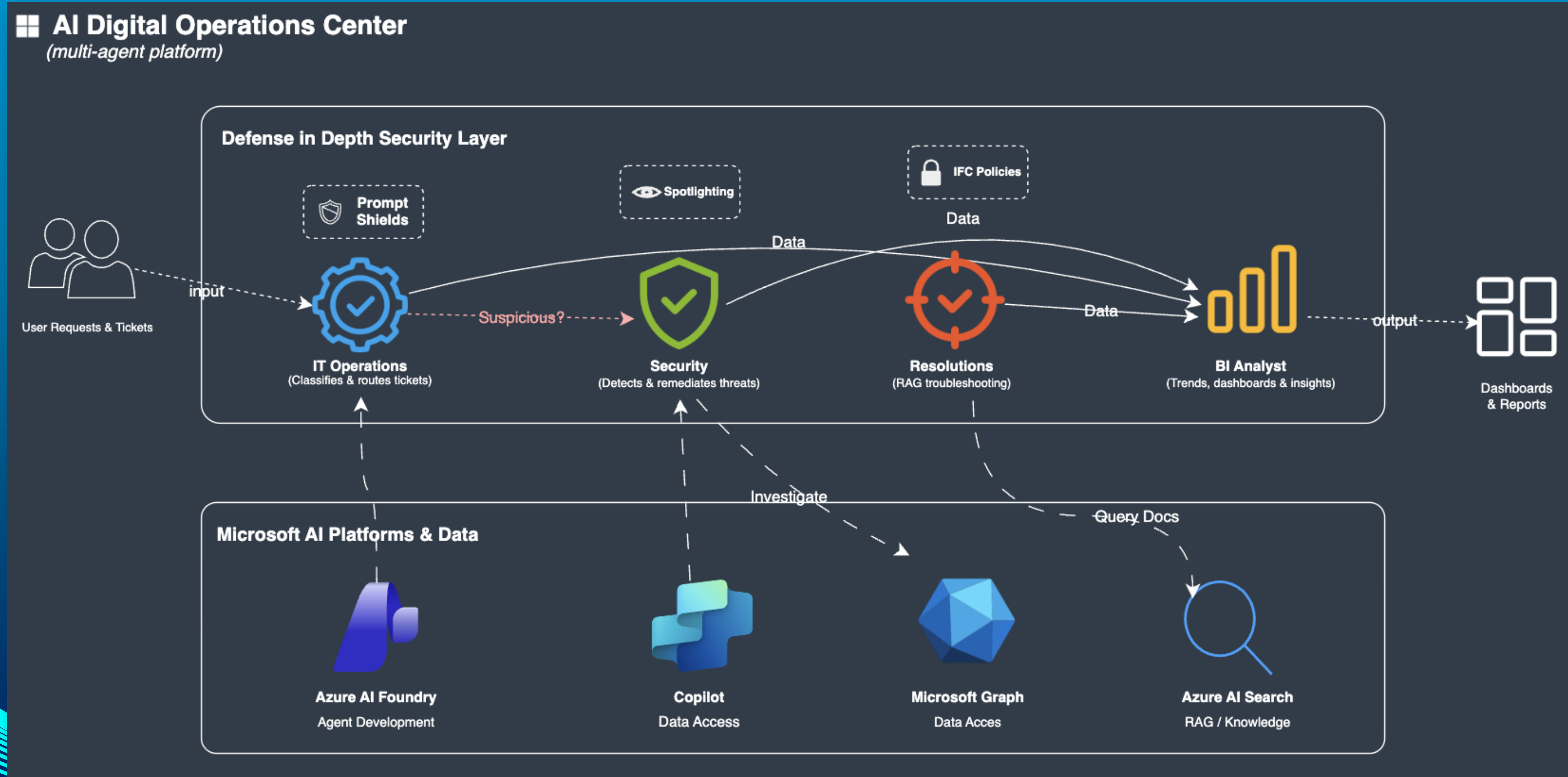
The Problem

- IT, Security and Business Operations teams work in separate teams
- Information is scattered across multiple tools
- Manual investigations increase response time
- Talent shortages make it difficult to keep up with incidents





My Solution: *A multi-agent system*



DEMO

Hello

 IT Operations Agent

Welcome to the AI Digital Operations Center! How can I help you today? I can help identify and route IT issues, assist with security concerns, provide troubleshooting guidance, and provide insights into IT operations.

We are experiencing a major IT outage across the company. Many Windows computers crashed and now show blue screens after a recent security software update. Employees cannot access important applications, and several teams cannot work. Can you look into this and help us restore operations?

 IT Operations Agent

Thank you for the details. Here's the assessment:

- **Category:** Software (Windows-related, linked to security software update)
- **Priority:** Critical (company-wide outage, multiple employees unable to work, significant business impact)
- **Issue Summary:** Multiple Windows computers crashing with blue screens after recent security software update, preventing access to essential applications and halting operations across teams.
- **Recommended Agent:** Resolution Agent
- **Reason for Routing:** Requires in-depth troubleshooting to identify the root cause of the blue screen errors and rollback or fix the problematic update. Coordinated response is needed to restore functionality quickly.
- **Relevant Context:** Issue began immediately  applying a security software update. No workaround appears to exist; operations are severely impacted.

+ Message IT Operations Agent





Benefits & Impact



Faster response

Incidents detected, triaged, and routed in minutes, not hours



Less manual work

Repetitive investigation and triage handled by agents



Connected teams

IT, security, and business operations share one view



Sharper decisions

Real-time dashboards turn operational data into insight



Summer Recap

This summer, I developed new skills by addressing real-world challenges and gaining hands-on experience with AI solutions, which advanced my professional growth in technology.

What I am taking with me

- Building a multi-agent Digital Operations Center from research to demo
- Learning how AI agents collaborate across IT, security, and business operations
- Connecting with mentors and teammates across Microsoft and the On Ramps program
- Growing more confident presenting technical ideas to a non-technical audience
- Next steps: earning cybersecurity certifications and continuing to build



THANK YOU

Any questions?